All right, good afternoon. Let's go ahead and recap our last lecture. So last time we were talking about the T distribution, so kind of another detooler talking about a new distribution that's useful for us. Kind of the main way we work with it through this generative definition. So if we have Z, that's a standard normal random variable, and Y is a kai squared random variable. Some degrees of freedom parameter are M, and these two things are independent, Z and Y. Then if we construct this ratio, Z over the square root of Y over its degrees of freedom M, as long as these conditions are met, then this new random variable T will follow a T distribution and it will have a degrees of freedom parameter M. We denote it with little T sub-M, T distribution with M degrees of freedom. We talked about its properties, right, its moments don't always exist. Degrees of freedom are large enough than they do. It has heavier tails than the normal distribution, but those heavy tails kind of decrease or subside as the degrees of freedom increases. And as that degrees of freedom parameter M goes to infinity, this T distribution converges to the standard normal distribution. Gave you the probability density function, but again, we don't typically want to integrate this. It's kind of a pain to integrate, so this would typically be done numerically. We're in our case using a T table. We will interact with T probabilities. Just looking at the probability density function here for different degrees of freedom parameters, And then kind of one of the main points here is that it has this relationship to normal sampling that we've been talking about so much. So if X1 through XN is a random sample from a normal distribution, mean parameter mu, variance parameter is the sample mean. We have this S squared sample variance, specifically dividing by n minus 1 here. If we construct this quantity, square root of n times x bar minus mu, over square root of s squared, the sample variance, This random variable U will follow a T distribution, and its degrees of freedom degrees of freedom parameter will be n minus one. Kind of one of the important points from last time's lecture relating it to the sampling distribution here. And then we just looked at some different examples, talked about how to use the T table. Today we're going to talk about interval estimation. So the only type of estimator we've really covered so far is point estimators. We've talked about different point estimators, Bayesian point estimators, Frequenest point estimators, the maximum likelihood estimator, two different estimators for the variance, but they're all point estimators. They all just give us kind of a single best guess of some parameter that we're interested in.

But again, our sample is random. We're going to have variation from our sample. And so point estimate is also going to be random. It's going to have variation. And it's not really ever going to equal the true parameter. It might be a good guess for the true parameter, but it won't be equal for the two parameter. And so oftentimes we want an interval estimate, some range of values that we think are reasonably likely for the parameter.

And that's what we're going to talk about today. So the basic idea is that we're going to instruct an interval now, and that interval is going to have both an lower and an upper bound.

And we want those lower and upper bounds to be likely to contain the parameter.

We're mainly going to talk about confidence intervals today, which is the frequentest interval estimate.

But you can construct Bayesian intervals. These are called credible intervals, and I'll just briefly mention those at the end. They're a lot more simple. It's pretty straightforward in the Bayesian case, so we're going to spend most of the lecture just talking about the frequentest case, confidence intervals here.

Okay, so definition, what is a confidence interval? Well, let's say we have some random sample again, X1 through XN, be from any population, whatever population you want, but that population has some parameters data, or maybe it's a vector of parameters, bold data, if you want. So we're going to talk about L and U, two statistics calculated from the data with L less than U. So you. So you can, you You can think of L being lower, U being upper. But there's two statistics

calculated from the data. The data is random, so L and you will also be random. You take a sample and take another sample, right? Both samples will give you different values for L and you because they're random. The data is random, so L and you are random. But if this probability inequality holds, if the probability that our parameter is between these random endpoints, L and you, if that probability is 1 minus alpha, then we call this interval L to you a confidence interval, specifically a 1 minus alpha confidence interval. If we set, for example, alpha to be 0.05, 1 minus alpha is 0.95, so it would be a 0.95 or 95% confidence interval. So it just has to meet this probability statement involving probabilities of these random intervals. L and you are random, right? So what's the probability that this fixed but unknown parameter theta falls between these random endpoints over repeated samples. That's the probability that we're talking. And we'll visualize this in a minute, but I want to go over a couple examples of conference intervals and how you construct them. And so as we've seen, we're very interested in the normal model, sampling from a normal population. So let's go ahead and look at intervals for the mean of a normal distribution, one of the most common parameters that we want to estimate. mean of a normal distribution. We've kind of seen two cases. We've seen the case where we know the population variance. We know sigma squared, but we don't know the mean. And we have seen the case, at least for point estimation, the more common case where we don't know either parameter, we don't know meo, and we don't know sigma squared. But let's start with the simple case. So we know the variance. So let's say we have a random sample, x1 through xN, IID from some normal distribution. And we know sigma, we know the standard deviation of the variance of this normal distribution. And if we want a 1 minus alpha confidence interval for the mean parameter, for mu, can be constructed in this way. X bar, the sample mean, plus or minus this extra term. So we have this constant z sub 1 minus alpha over 2 and then times our standard deviation sigma and then divided by the square root of n. So we see this second term or what's called a margin of error. We see this gets smaller as n gets bigger, dividing by a bigger number. So the margin of error gets smaller. Our interval will get smaller as we get a bigger sample size. That should make intuitive sense. If we collect more data, we're less uncertain. We're more certain about the parameter. So as we collect more and more data, we would expect to have narrow error intervals. And then this constant here, Z, 1 minus alpha over 2, this is just the quantile of a standard normal distribution, normal distribution with mean zero and variance 1. And specifically, it's the 1 minus alpha over 2 quantile of that standard normal distribution. So it's x bar, our sample mean, or our point estimate for the mean, plus or minus some margin of error. And that margin of error has three ingredients. It has the standard normal quantile. It has the currently known population standard deviation. And it involves the sample size end. This is a theorem I'm giving it to you. You can use this now whenever it's applicable. So if you're working on a problem and you have a random sample from a normal distribution where you know the variance and you're asked to compute an interval estimate, a confidence interval, you can go ahead and use this directly. Let's go ahead and show this here. You're going to see a common theme when we calculate confidence intervals. So we're always going to start off with some distribution that we know. So in this case, I'm going to look at what I'm going to call Z here. It's calculated as square root of n times x bar minus mu over square of sigma squared. Who remembers what distribution z follows? We're plugging in the true variance here in the denominator, not the estimated variance. What was that? not kai squared, standard normal. So this is your first homework problem, I think, for today. But this is the square root of n times x bar minus mu over the square root of sigma squared, the true sigma squared. This will follow a standard normal distribution, normal 0.1. So we could make a probability statement about a standard normal random variable. That's easy. We know it's mean. We know it's variance. There's nothing unknown about a standard normal random variable. And so if we drew a standard normal probability density, something like this, the middle is zero.

And if we want one minus alpha probability in the middle, that means we want alpha over two probability tail, each of these shaded tails, alpha over two probability.

We could look up what these values are in our standard normal table or our T table at degrees of freedom.

This one to the right, we have a table that gives the area to the left. So this is going to be Z to the one minus alpha over two.

And since the normal distribution is symmetric, we know that this. this and this are just sign swapped. We're reflecting about zero. So this value on the other side is negative z 1 minus alpha over 2. If we know something is normally distributed, we can find these values, Z 1 minus alpha over 2, negative z 1 minus alpha over 2, such that this green area is 1 minus alpha. 1 minus alpha probability in the middle. We could do that with our normal table. And we know that this quantity follows a normal distribution, standard normal. So that's kind of a strategy. We could write the probability that negative, z 1 minus alpha over 2 is less than or equal to the random variable z that follows a standard normal distribution less than or equal to little z 1 minus over 2 and that probability is 1 minus alpha z is standard normal we would all agree with this right there's nothing unknown that's just coming straight from the table we all agree with this and now let's just plug in our expression for z is equal to probability negative little z 1 minus alpha over 2 is less than equal to square root of n x bar minus mu over square root sigma squared and that's less than or equal to positive z 1 minus alpha over 2 and that's equal to 1 over alpha 1 minus alpha so we started with something whose distribution did not depend on the parameter this was calculated or this is a function of the parameter but its distribution is standard normal Its distribution does not depend on the parameter. Since its distribution does not depend on the parameter, we can construct this probability statement. We can find this lower bound and this upper bound, using our table, in this case, a normal table. And then we're just plugging in our expression for z here. Now, if we want an interval for mu, we want something of this form. In this form, we have 1 minus alpha on the right. We already have that. And then we have an inequality in the probability statement, and in the middle of that inequality is the parameter. theta.

So we want that same setup. We want the parameter, in our case, mu, in the middle of our inequality. Right now, we don't have mea, we have square root of n times x bar minus mu over square root of sigma squared. But we have an inequality, so we can just solve for mu, and that's the idea. So we're just going to start rearranging this inequality, solve for mu, until we get something that's of this form. And that's the same recipe we're going to use for other confidence intervals. The distributions that we work with might be different, we're always going to take a probability inequality and rearrange it, solve for our parameter in the middle. Okay, so let's go ahead and see that here. So we have square root of n times x bar minus mu over sigma over square of sigma squared. So let's start by multiplying all sides of the inequality by square root of sigma squared, dividing by square of n. So then we would have the probability that negative z minus alpha over two times square of sigma squared or square of n. this is less than equal to x bar minus mu that's less than equal to z 1 minus alpha over 2 times square of sigma squared or squared of n and again that's still equal to 1 minus alpha but again we want mu in the middle so we're going to have to subtract x bar from all sides of the inequality and then multiply by negative 1 because right now we have a negative mu and when we multiply an inequality by a negative number we're going to flip the signs so rearranging in this last step We're going to have probability x bar plus z 1 minus alpha over 2. We flip the sign of our inequality. This is greater than equal to mu the sign. This is greater than equal to x bar minus z 1 minus alpha over 2, square root of sigma squared over squared of n. Times square root of sigma squared over squared of n. And again, this probability is still as alpha, and this is the same, the probability that x bar minus z, 1 minus. alpha over 2, square of sigma squared over squared of n, less than or equal to mu, less than or equal to x bar, plus this margin of error, z, 1 minus alpha over 2, square of sigma squared or square of n, and that's 1 minus alpha. And this now is in the same form as our definition for a confidence interval. We have our parameter of interest in the middle, we have a lower bound and an upper bound. They're both random, their functions of the random data, and that probability that the

parameter, the fixed but unknown parameter mu is between these random lower and upper bounds, that probability is 1 minus l. This interval, x bar, plus or minus, our margin of error, z 1 minus alpha over 2 times sigma over the square of n for our parameter mu. We started with something that was a function of the parameter, a transformation of it in some way involving the data, and it had a distribution, standard normal in this case, a distribution that didn't depend on the parameter. The standard normal distribution does not depend on mu, or another way of saying this is this holds regardless of what the value of mu is. As it holds for any mu, we can go ahead and find these bounds for the standard normal, negative z1 myself over 2, z1 myself over 2, plug in our expression for z and then solve for the parameter.

It's the idea. Should have gone down here. Now let's do an example with hypothetical data. So let's say x1 through xN is a random sample of daily rainfall measurements, for Santa Cruz in the month of February, the sample size N of 26. After collecting the data, we calculate the sample mean X bar, and it's 1.3. And let's say we know that rainfall in February is normally distributed, and we know the variance, let's say it's one fourth, but we don't know what the population mean average rainfall is. We don't know a mu. Let's use this information, instruct a 95% confidence interval for the average daily rainfall. Start by finding our value of alpha. We know 1 minus alpha 0.95 because we've been requested to make a 95% confidence interval. So alpha. Since we know this is normal, we can use this theorem here directly. So we know we're going to need this constant here, this quantile of the standard normal distribution. The 1 minus alpha over 2 quantile of the standard normal distribution. So if alpha 0.05, then z 1 minus alpha over 2 is just z 0.975. We need to find 0.975 quantile, the standard normal distribution. Where can we find that?

T table. So if we go to our last lecture, the T distribution, at the very end, we had degrees of freedom, infinity, this last row, and that's our standard normal distribution, this last row. And we're looking for the 0.975 quantile. So this column or this value here, 1.96. So Z, 0.975, 1.96. So we know probability that x bar, our sample mean, minus quantile of the standard normal distribution, 1.96, times the known population standard deviation sigma over the square root of n, less than or equal to mu, and or equal to x bar plus our margin of error, 1.96 sigma, the square of n. This probability is what?

0.95, or 1 minus alpha. Or this is the same instructing the interval, x bar plus or minus, quantum. 96 and sigma over the square of n.

tile 1. And we have these values.

We were given x bar equal to 3.

1. Sigma is the standard deviation. We were given the variance. The standard deviation is the square to the variance or the square root and fourth in our case is just one half.

And we were given the sample size n equal to 26. So we have all of the ingredients we need to construct this confidence interval.

1.3x bar plus or minus our quantile from our standard normal distribution. 1.96, 1 half over the square root of n squared of 26, rewriting this as 3 plus or minus, 1.96 over 2 root 26.

This is a 95% confidence interval. I'll just write CI for our unknown mean parameter. So because we identified that it's a normal population, random sample from a normal population, and we had the true variance one-fourth, we were able to just use this theorem, construct our confidence interval in this way. Questions here. Okay, so that's the known variance case when we know the population variance, sigma squared. Most often in practice, we won't actually know that. So let's look at the unknown variance case. One of the most frequently occurring statistical problems, say we have a random sample, X1 through XN, coming from a normal distribution with unknown mean mu and unknown

variant sigma squared. And let's say we're still interested in constructing a confidence interval for mu, specifically a one minus alpha confidence interval from you. In this case, where we don't know the true population sigma squared, we can estimate it. Estimate it using our sample variance, s squared, that we've talked about. You see s here instead of sigma. The other thing that changes is we no longer have a quantile of a z distribution, a standard normal distribution, we have this t quantile. This stands for the 1 minus alpha over 2 quantile of a t distribution, where that t distribution has n minus 1 degrees of freedom. So in this case where the population variance is unknown, the overall form of our interval estimate is very similar. It's still x bar, our point estimate, plus or minus, a margin of error. But now that margin of error uses a quantile from a T distribution, and we're estimating the variance or estimating the standard deviation instead of plugging in a known value. So again, this T is just the 1 minus alpha over 2 quantile of the T distribution with n minus 1 degrees of freedom. So you'll figure out what n minus 1 is, and you'll go to the corresponding row of your T table to look that value up. Yeah. We're going to go over that in just a moment. It's part of the proof here. And again, S is our estimated standard deviation, the square root of the sample variance that we're calculating with this N-minus-1 denominator. Right, if you remember in our lecture last time on the T distribution, right, we had to use this denominator, N-minus 1 to get that T result. So we're going to do that here as well. Okay. So why is it a T and not a standard normal quantity? We're going to follow sort of the same strategy. We're going to start with something that we know the distribution of, a distribution that does not depend on mu. I'm going to look at this random variable. I'm going to call u, instructed as square root of n, x bar, our sample mean, minus mu, over square root of s squared now. We don't know the true variance. We don't know sigma squared. If we looked at the known variance case, we started with something similar, but we had sigma squared, the known. variance in the denominator. Now we don't know that, so we instead plug in s squared, the sample variance, who remembers the distribution of this, T distribution with n minus one degrees of freedom. Yeah, that was from our last lecture. So this quantity now will follow a T distribution, n minus one degrees of freedom. Have some construction that involves our data and involves the parameter mu, but the distribution of that construction does not involve the parameter.

T n minus one, distribution does not involve mu. Or again, another way of thinking about this is that this result holds for any mu, holds regardless of the value of mu. So we're going to start there. We're going to construct probability inequality, and we're going to solve for mu just like we did in the known variance case. Here, again, the T distribution is symmetric, the standard normal, so we know the probability that negative t n minus 1, comma, 1 minus alpha over 2, less than or equal to this random variable U that follows a T distribution, with n minus 1 degrees of freedom, less than or equal to positive quantile of this T distribution, T n minus 1, 1,000, 1 minus alpha over 2. This probability is what? 1 minus alpha. So this is just notation that means the 1 minus alpha over 2 quantile of a T distribution with n minus 1 degrees of freedom. And just like our little diagram for the standard normal case, since it's symmetric, right, we know the area in the middle is 1 minus alpha, and we get that a positive value, flipped sign of that value.

Same thing here. We're just using a T distribution instead of a standard normal. So we have a T quantile instead of a standard normal quantile. We all agree with this probability statement.

And let's go ahead and plug in our expression for you now. So this is the same as saying probability that negative t n minus 1, number 1 minus alpha over 2 is less than or equal to square root of n, x bar as mu, square root of s squared, less than or equal to a positive quantile t n minus 1, 1 minus alpha over 2, and this probability is 1 minus alpha. And now we have probability inequality. We want to isolate the parameter in the middle, right? We want mu in the middle of this inequality, so we can go ahead and solve. Multiply by square root of s squared, divide by square root of n, subtract x bar, and then multiply by negative 1. Flip the inequalities solve, just like we did in the standard normal

case. And then we get the probability x bar minus, this t quantite. t n minus 1, n minus alpha over 2, square of s squared, squared of n, less than or equal to the parameter of mu, than are equal to x bar plus a margin of error, t quantile n minus 1, 1 minus alpha over 2, square root of s squared over squared of n.

This probability is 1 minus alpha, and this is the same form as our definition of a confidence interval.

A probability that a random lower bound, less than or equal to the parameter, less than a random upper bound, is equal to 1 minus alpha. This interval, x bar, plus or minus, our margin of error, which is this T quantile now, TN minus 1, comma, 1 minus alpha over 2, S over the square of n is for our parameter meal. Okay, so let's look at a data example now. Suppose we have a random sample, X1 through XN, again, of daily rainfall measurements for Santa Cruz in February. We have the same sample size n equals 26. We get the sample mean x bar, it's still 1.3. But now let's say we estimate the sample variance, and we get an estimate of 0.45. We don't know the true variance, but we have this estimate of the variance. Again, we know these are normally distributed. Use this information to construct a 95% confidence interval for the parameter of mu, the mean. What do we know? We have the sample size n26. That's one ingredient from our theorem here. We need x bar, this t quantile, we need s, and we need n. Those are the ingredients to this confidence interval for the parameter of mu. We've identified it's normally distributed, it's a random sample from a normal distribution, creating a confidence interval for the mean, and we don't know the variance, so we can use the above theorem. So we know n is equal to 26. We know s squared, our sample variance, estimated at 0.45, I had to take the square root of that. since we need s, s is the square root of 0.45, 7, x bar, 3.

or about 0. And then we need our quantile from our T distribution.

T, n minus 1, comma, 1 minus alpha over 2.

N minus 1 is 26 minus 1. 975.

5.1 minus alpha over 2, 0. So we want the 97.5 quantile of a T distribution, 25 degrees of freedom. We can go ahead and look that up in our T-table, me five degrees of freedom.

So this row here, 0.975 quantile, 06.

so this column here, and we get this value 2. That's the quantile 06.

for our confidence interval is equal to 2. And now let's go ahead and just construct our confidence interval, just like in the theorem. Right, x bar plus or minus our t quantile times s over the square of n expression here.

X bar plus or minus n minus and minus one comma one minus for two, s square root of n, it's 1.3 plus or minus, 2.06, 0.67 over the square root of 26. And this is a 95% confidence interval, CI, or unknown mean, mu.

Again, I've given you two theorems now related to normal sampling. So if you identify that you have a random sample from a normal population and you want a confidence interval for the mean mu, you can use the corresponding theorem depending on whether or not you know the true variance or you're estimating it. Some confidence intervals will use a kye squared distribution.

We'll look at an example in a minute.

In these cases, it's strictly because we know that this expression involving mu follows a t distribution, so we're going to use t quantiles. In this case, it was standard normal, so we're going to use standard normal quant. But it could involve a kai squared random variable, and it could be something that's not symmetric.

In these cases, these distributions, standard normal and t are symmetric, so our interval is our point estimate plus or minus a margin of error. The point estimate falls right in the middle. That doesn't always have to be the case. It worked out nicely in these cases that it did.

Other questions here?

So I want to talk about the interpretation of confidence intervals. This is probably the trickiest part about confidence intervals. So our definition involved this statement here.

It said the probability that L less than Theta less than U is 1 minus L.

L. L and U here are random variables. They inherit their randomness from the data. They're calculated based on the data. The data is random, so L and you are random. So this is a statement about random variables. Theta is not random. We're being frequentists. So theta is not random. It's fixed but unknown. L and U are random, lower, and upper bound. If I took a sample and computed a lower and upper bound, and I took another sample, computed the lower and upper bound, they would be different. I did this many, many, many times. I'd get a different value each time, or I get a different interval each time. interval itself is random.

Now once we collect data and we compute the specific lower and upper bounds for those data, it's just say capital L is equal to little L. Little L is no longer random. Capital U is equal to little U, little U, little U is no longer random. Well, then they're no longer random. So in that case, little L would not be random, theta is not random, and little you would not be random. So after we compute a specific interval, we're not talking about anything random.

So we can't make probability statements.

Probability involve random things, and once we've computed a specific interval, it's no longer random. The interval either contains the parameter, or it doesn't. There's no randomness involved, no probability involved.

So what we do instead is we interpret things in this long-run frequency, this idea sampling distribution, right?

If we sample again and again and again and compute many of these intervals, it's not something we could feasibly do in practice, but we can understand the longer and probability of that procedure.

And so that's how we interpret. If we were able to take mini samples and for each of those many samples compute their confidence intervals, then we would expect about 1 minus alpha percent of those to cover the true parameter. This is a statement about the procedure of constructing these confidence intervals. It's not a statement about any particular confidence interval that you get for a single data set. We don't make probability statements for any particular interval from a specific data set. I put together a little simulation to kind of visualize this. So I'm going to take repeated samples, not something you would do in practice, but you can do in a simulation. So I'm going to take repeated samples. Each sample will have size 26, and they'll be sampled from a normal distribution with mean 5.1 and standard deviation 1.6. I'll take 26 observations randomly from this distribution, and I'm going to compute a confidence interval, and I'm going to do that many times. I'm going to do it 100 times. Be specific. For each of these 100 samples of size 26, I'm going to compute a confidence interval. And I know the true mean. It's 5.1, so I can compare whether or not those confidence intervals contained the true mean. And this is what this looks like. Each kind of row going down is one single confidence interval. So in this case, value 4.75 to just over 6. That's this first confidence interval. The next one's a little different. The next one's a little different. And this dashed line going down the middle. is the true mean.

And then I've colored all of these intervals blue if they happened to cover the true value and I've covered them red if they did not.

So if we just look at this first row, what's the probability that this first row covers the true value?

It's one.

What's the probability that this red one here covers the true value?

That's why we say for any given interval probability statements don't really make sense. It either covers the interval or it doesn't. Now in practice we wouldn't have this dashed line so we wouldn't know if it was 1 or 0, but it would be 1 or 0. We can't make probability statements about any given interval collected from a data set. What we can see is that if we were to repeat many random samples and construct intervals, roughly 95% of them would cover the true value. In this case, there's 7 that don't, 93 that do, pretty close to 95 out of 100. That's the probability statement we're making. It's over repeated sampling. it's not for any given interval from a single data set. So if you look at this one, for example, alpha gets smaller. It means you want a bigger confidence interval. This T quantile gets bigger. It means we're going further to the right in our T table, right? And those values get bigger as we go further to the right. So that's what's causing the interval to get bigger. Our margin of error gets bigger, specifically because this quantile gets bigger. S, square root of n, and x bar might not change. But if you want a 90-first. conference interval, that's going to be smaller than a 95% confidence interval, because the 95.

quantile for 97.5 is different than 0.95. So are we all somewhat comfortable with this interpretation of probability for conference intervals? Again, we're not making any probability statements about a single interval, right?

So if you go out and you calculate a conference interval for some data that you collected, you don't say there's a 95% probability that the interval covers the true value. The probability is 0 or 1. You don't know. but you say you're 95% confident, which is really this long run repeated sampling interpretation of a confidence.

Sure, if I give you the interval negative infinity to infinity, you could probably construct special cases, but yeah, it's not typically something you would do.

Yeah, if I have a probability parameter, I know the probability is between zero and one because it's a probability, so I can give you an interval of zero to one. It has finite bounds. Not a particularly useful interval, though.

I don't know. It sort of gets not. very useful, right? If you, you can always construct a bigger and bigger interval, but if it's such a wide range that there's no practical use, the question would be why. So it needs to be practically.

Now, maybe if you had a ridiculously high sample size, like billions, then maybe you could construct, you know, a 99.9% interval that's not overly wide for whatever your purposes are. But that would be kind of the exception, not the rule. It's like, it's simple.

Like, what is that giving you? I'm not sure I followed, but one thing I'll note here is that the sample size was the same in every case. So in each of these, the sample size was 26. We constructed our interval using 26 observations. So that didn't change from one to the next. That was always the same. We just repeated this process of taking 26 observations and constructing an interval. We repeated that process many times, specifically 100 times in this. Does that answer the question? So in the 50% compared to the 95, the only thing that's changing is this quantile. This quantile is going to be smaller, so those intervals are going to get smaller. So maybe, you know, maybe this interval goes

from here to here, and this one goes from here to here. And so you can see if you shorten all of those intervals, the number of them that cover is going to drop. Specifically, if you shorten them so that it's a 50% interval, 50% of these, on average, will cover the true value. I'm not sure what you mean by condensing the sample findings. We're not cutting off observations, right? We have a summary. X bar is a summary. It uses all of our observations. S is a summary. It uses all of our observations. So the only thing that's changing is T, but T doesn't involve the data. T is based on our distribution that's sort of independent of the parameter. X bar and S are going to be the same regardless of which T we choose. So we're always going to use all of the data when we construct it. it's scaling this term specific. It's not scaling x-bar, but x-bar is the middle point. So it's the middle point, plus or minus a margin of error, right? The margin of error involves this s, but it also involves this quantile term. Okay, so that's the interpretation. Again, I've kind of said this a couple times, but notice that when we're constructing these confidence intervals for this parameter in mu, coming from a normal population, we used one of these two facts. So in the case where sigma squared was known, we used the fact that this construction, sigma squared or squared of n times x bar minus mu over sigma, this followed a standard normal distribution.

In case where cina squared was not known, right, we can plug in an estimate. We can plug in our sample variance into the denominator. Again, in that case, it was a t distribution with n minus 1 degrees of free. So in each of these cases, we had some construction involving the parameter of interest, mu, and involving the data as well.

But the distribution of that construction did not involve.

That's important. We've been using our sample distribution results to get here. These are both things that we've established from the last few lectures on sampling distributions. So we've sort of been working up to this point. Now that we understand sampling distributions, we understand the distribution of these things, we see that these distributions are independent of the parameter, and that's what allowed us to get confidence interval.

And so this strategy works in many other cases. And so the idea here is let's say we have some random sample X, X1 through XN, from a population that depends on some parameter, theta. In our cases that we looked at, it was mu, but it could be whatever parameter, theta, or vector parameters.

And let's say we have this function of Q that involves the data X, but also the parameter.

This was one function Q. This was one function Q. It involves the data and the parameter. It's just a function that takes in the data, takes in the parameter.

But the distribution, Q does not depend on theta. Again, another way I've been saying, this is the distribution's the same, regardless of the value the parameter of theta.

If we have something like that, like we did in these two cases, if we have something like that, then this function Q is called a pivotal quantity.

This was a pivotal quantity.

This was a pivotal quantity. It's a function of the data and the parameter, and the distribution of that function does not involve the price. So this is nice because if you can find a pivotal quantity that involves the parameter that you're trying to get an interval for specifically, right?

In this case, it had to be a pivotal quantity involving mu, because we're constructing confidence interval from you. We don't want a pivotal quantity for some other parameter. But if you can find one for the parameter that you're interested in, then this gives you a way to construct a confidence interval. And

we've already kind of seen it. This general recipe is find some value C1 and some value C2, such that we can have this probability and a probability.

The pivotal quantity is between some lower bound and some upper bound, and that probability should be 1 minus alpha.

So if we can find such a C1 and such a which is C2, right?

Those are quantiles of the distribution of this pivotal quantity Q. It should be unknown.

Okay, so we can find these quantiles, C1 and C2, but give the probability 1 minus alpha being between those two values for the pivotal quantity. And then we just need to isolate the parameter from that pivotal quantity. We saw two examples of that. Basically, we're just finding some function G, such that when we apply G to the pivotal quantity, we get back theta.

We isolate the parameter of theta. In our case, as we can, got it by multiplying by square of n or dividing by square of n, multiplying by either s or sigma, subtracting mu or x bar, and then multiplying by negative one. Right. We just need some function that's going to give us theta, isolate theta, when we apply it to Q.

And then we can take a probability inequality, apply this function G to all sides of it.

We're just solving for the parameter in the middle, solving for the parameter in the middle. So some function of the lower bound, less than are equal to the parameter, less than are equal to the function of the upper bound. And that probability is still 1 minus alpha. If we can do that, you have a way to get an interval. This is our lower bound. This is our upper bound for the parameter theta. Now these constants, C1 and C2, that you can choose, they're not unique. There's really any number, infinite number, that you can choose. But what we typically do is we'll set C1 to be the alpha minus 2 quantile, and we'll set C2, the upper one to be the 1 minus alpha over 2 quantile. If we go back to a picture here, this is the 1 minus alpha over 2 quantile. This is the alpha over 2 quantile. We did that so that there was the same probability in each tail, alpha over 2 in each tail. So that's one choice that you could make. There's an infinite number, but that's one choice you could make. And this is what's called an equal-tail interval. This is what you would typically do unless you had a strong reason to do otherwise. So that's the general strategy for finding a confidence interval. First, find a pivotal quantity, function of the data and the parameter whose distribution does not depend on the parameter. Instruct a probability inequality, solve for the parameter. That's the idea. So let's go ahead and do one more example that's not for the mean, right? So we can see this kind of rule in action. So let's say we have data x1 through xN that forms a random sample from the normal distribution, unknown mean mu, and unknown variance sigma squared. And now we want a 1 minus alpha composite interval, not for the mean, not for mu, but for the variance. Can anyone think of a pivotal quantity for the variance from a normal distribution? Back to our joint distribution of the sample mean and variance lecture. A pivotal quantity for the variance. Function of the data and the variance. A function of the data and the variance whose distribution does not depend on a meter sigma squared. So this is the piece I'm looking for. the function of the data and the parameter. And it will follow some distribution. So I'm not looking for a distribution yet. I'm just looking for a function of the data and the parameter. Yeah. Gamma distribution, it's going to be harder to find the quantiles. What about something related to the gamma distribution? It's a function of the data and the parameter of interest, in this case, sigma squared. So it's sigma squared hat over sigma squared, which is the same as n minus 1 as squared, over sigma squared, right, and that thing follows a kai squared distribution with minus one degrees or three. So we have pivotal quantity.

We have something that's a function of the data and a function of the parameter

of interest, single squared, and the distribution of this quantity does not depend on sigma squared. It's the same regardless of sigma squared. It's always kai squared and minus one. So we have a pivotal quantity.

We go ahead and use this now. So working with a kai square distribution, it's skewed, it's not symmetric.

Maybe it looks something like this. Let's go ahead. ahead and do an equal tailed confidence interval.

So we want alpha over two probability in each of these tails.

We know this value is going to be alpha over two quantile.

This value is going to be the one minus alpha over two quant. Because our table gives us the value to the left of a point. Yeah. This is s squared, yeah.

So we know the probability that this lower quantile, kai squared, say n minus one degrees of freedom, alpha over two, less than or equal to. This is a number. This random variable y, follow the kye squared distribution with n minus 1 degrees of freedom, less than or equal to upper quantile, kai squared n minus 1, 1, minus alpha over 2.

We know that this probability is 1 minus alpha.

Let's go ahead and plug in our expression for the pivotal quantity.

The same as saying probability is lower quantile, kai squared n minus 1 or 2, then are equal to n minus 1, s squared We're sigma squared less than are equal to upper quantile, kye squared n minus 1, minus alpha over 2. This probability is also 1 minus alpha. And then we're just going to start solving to get our parameter in the middle, sigma squared in the middle. First step is to invert all sides of this inequality. So if we invert, we're going to have to flip the signs. Okay, so this is the same saying probability 1 over this quantile. kye squared n minus 1 comma alpha over 2 less or greater than equal to sigma squared for n minus 1 s squared greater than equal to 1 over the other quantile i squared n minus 1 comma 1 minus alpha over 2 probability is still 1 minus alpha and now multiply all sides by this denominator n minus 1 over s squared so we have the probability that n minus 1 s squared for this quantile kai squared n minus 1 and minus 1 alpha over 2.

This is greater than or equal to sigma squared, greater than equal to n minus 1, s squared over the other quantile, chi squared n minus 1, comma 1 minus alpha over 2.

Or just rearranging this, this is the probability, n minus 1, s squared, upper quantile now, kai squared, n minus 1, minus alpha over 2, less than or equal to sigma squared, if n or equal to n minus 1, s squared over the lower quantile kye squared n minus 1 comma alpha over 2 and the probability is still 1 minus alpha so we have probability statement of probability inequality with our parameter in the middle a lower round upper bound both random and that's equal to 1 minus alpha that meets our definition of a conference interval so this interval n minus 1 s squared kai squared n minus 1 comma 2 as the lower bound n minus 1 S squared, kai squared, n minus 1, alpha over 2 is the upper bound. This interval is a 1 minus alpha, competence interval, parameter, same with squared now, population variance. Yeah, so it's not symmetric, it's not centered around zero, and you flip it, so we need two different quantiles here. It's not like, in the normal in the normal in T cases, it was two quantiles, but one was just the negative of the other, so you only have to look up one. Here, one's not the negative of the other, so we have to look up both, but it's the same idea, right, we just take the distribution, we find the alpha over two quantile, and the 1 minus alpha over 2 quant. It just so happened that in the standard normal and T cases, the alpha over 2 quantile was negative 1 times the 1 minus alpha over 2 quant, due to the symmetry around zero for those distributions.

So let's go ahead and see this with some data now.

We have a sample X1 through XN, random sample of these rainfall measurements. The size N is still 26.

After we collect the data. We calculate x bars 1.3. We calculate the 45.

sample variance. We know he's come from a normal population.

Now construct a 95% confidence interval for the variance, the population variance, daily range.

So we know n26. We know the sample 45.

variance has squared is 0. We have a 95% confidence 05.

interval, so alpha is 0. And so we're going to need two quantiles.

We're going to need for the kai squared distribution with n minus 1 or 25 degrees of freedom. We're going to need the alpha over 2 quantile, 0.05. And we're going to need kai squared 25 degrees of freedom, 1 minus alpha over 2, or 0.975. We're going to need both these quantiles, but this distribution does not depend on sigma squared. So we could look these up in our kai squared table. So we have 25 degrees of freedom, and we're looking for the 0.025 and 0.975 quantiles. We go to our kai squared table. Here, so we said, 025, the lower quantile is 13.12. And then we need to go to the next page. Here, and we wanted the 975 quantile. So we see 40.65. That's our upper quantile. And so going back to the problem, the lower quantile, the 0.025 quantile was 13.12. And the upper 97.5 quantile was 40.65. and we said this interval here was a confidence interval, right? So we know n, we know s squared, and we know these quantiles now. We can go ahead and plug that in. 25 sample size minus 1 times s squared, 0.45 over the upper quantile, 40.x5. That's our lower bound, 25 or n minus 1 times s squared or 0.45, divided by the lower quantile, 13.12. That's our upper bound. And if you plug that into a calculator, you get about .277 to 0.857. This is a 95% confidence interval for the variance parameter for sigma squared. So I've given you two theorems for very frequent confidence intervals, both involving the normal distribution and estimating the mean parameter, one where the variance is known, one of the variance is unknown, so feel free to use those theorems. but also be familiar with this sort of general strategy of taking a pivotal quantity, writing down a probability statement, probability inequality involving that pivotal quantity, and solving or isolating your parameter of interest in the middle of that inequality, just rearranging it to get a conference interval, just like we did in this last example here. So that's conference intervals. I'll just briefly mention the Bayesian version of interval estimation. And so putting our Bayesian hats back on, it's been a while, but in the Bayesian sense, setting, treating the parameters as random variables rather than fixed and unknown quantities. And we have some distribution that expresses our state of knowledge about the parameter in the given time. Usually we have a prior distribution before we collect the data. We have a posterior distribution that updates our beliefs after we've collected the data. We get there through Bayes rule. In the Bayesian setting, it's pretty straightforward to get an interval estimate because we have a distribution. We have a distribution tells us our state of knowledge about the parameters. that distribution and make a probability interval, probability 1 minus alpha. That's the idea. We just work directly with the posterior distribution, construct an interval from that distribution that has probability 1 minus alpha. Yep. So in the Bayesian case, if you construct, it's called a Bayesian credible interval, then you can say the probability that the parameter is in that interval is 1 minus alpha. We don't, we're not attached to this long run frequency interpretation of probability anymore. We're treating the parameters as random variables. now. They're not fixed so we can actually say this probability is one minus alpha for a specific interval that we've constructed. So that's nice. It's much more intuitive, much easier to understand what's going on. So we can get this directly from the

posterior. It's called a credible interval. Let's say we have data from a random sample or theta.

We find values L and U, such that the probability that the parameters between L and you, given the data X or using our posterior distribution, if that's 1 minus alpha, then the interval L to U called a one minus alpha credible rather than confidence interval. So this probability we can get just by integrating posterior distribution. So the integral from L to U, the lower bound to the upper bound, of that posterior with respect to the parameter, is equal to 1 minus L. If we can find an L and a U that give us that probability, then that interval L to you is a credible interval. And just like in the confidence interval case, right, these bounds L and U aren't unique. You could shift them up or down and get the same probability. So again, you'll very often work with an equal tail credible interval. So you'll put alpha minus two probability below L, and you'll put alpha minus two probability above you. Sort of likely do. Any questions on that? Do this one? Okay. Well, in there for the day, and I will see you on Thursday. I'll stick around for a bit if there's questions.